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Chapter 1: The Magic of Electricity and How to Save It

One fine morning, Ms. Priya walked into the classroom with a big smile. She had a torch in one hand and a fan in the other. The students were curious.

"Good morning, everyone! Today, we will talk about something that makes our lives easier every day—electricity! But do you know how it is made and why we should save it?" she asked.

Aarav raised his hand excitedly. "I know! We use switches to turn on lights and fans!"

Ms. Priya chuckled. "Yes, Aarav! But where does electricity come from? That's what we'll discover today."

1.1 How is Electricity Made?

Maya tilted her head. "I think power plants make it. But how?"

"Great question, Maya! Let me tell you a story to explain," Ms. Priya said.

"Imagine you are riding a bicycle. When you pedal, the wheels move, right? Now, think of a big water wheel near a river. When water flows, the wheel turns. If we connect the wheel to a machine called a generator, it can make electricity! This is how hydroelectric power plants work."

"Wow! So water makes electricity?" Anaya asked.

"Yes! But not just water. There are many ways to generate electricity. Let's explore them."

1.2 Different Ways to Generate Electricity

1. Thermal Power Plants (Using Coal or Gas)

Sam was curious. "What about those big factories with chimneys?"

Ms. Priya nodded. "Those are thermal power plants. They burn coal or gas to make heat. The heat boils water to create steam. The steam turns a big wheel called a turbine, which is connected to a generator that makes electricity."



"That sounds powerful!" Aarav said.

"It is! But burning coal and gas creates smoke and pollution. That's why we need better ways to make electricity."

2. Hydroelectric Power (Using Water)

Maya smiled. "Oh! Like the waterwheel in your story!"

"Exactly, Maya! Dams store water, and when water is released, it turns turbines to generate electricity. It's clean and doesn't pollute the air."

3. Solar Power (Using the Sun)

"I have seen solar panels on rooftops!" Anaya said.

"Good observation! Solar panels absorb sunlight and turn it into electricity. It's clean and unlimited, but it only works when the sun is shining."

4. Wind Power (Using Wind)

Sam jumped in. "I saw windmills when we visited the countryside! Do they make electricity too?"

"Yes, Sam! Wind turbines work like fans, but instead of using electricity to spin, they use wind to turn and generate power."

5. Nuclear Power (Using Atoms)

Aarav scratched his head. "What about nuclear power?"

"Good question! Nuclear power plants use tiny particles inside atoms to create a lot of heat, which turns water into steam, spins turbines, and generates electricity. It's powerful but needs to be handled carefully."



1.3 Simple Generator: Making Electricity at Home

Ms. Priya smiled. "Want to try making electricity yourselves?"

The students nodded excitedly.

"Take a magnet and a coil of wire. When you move the magnet inside the coil, you create an electric current! This is the basic idea behind generators."

Sam's eyes widened. "That's like magic!"

"It's science!" Ms. Priya laughed.

1.4 Renewable vs. Non-Renewable Energy

"We talked about different energy sources. Some last forever, and some don't," she said. "Can you tell me which ones are renewable?"

Maya raised her hand. "Solar, wind, and hydro!"

"Perfect! And non-renewable ones?"

Aarav replied, "Coal, gas, and nuclear."

"Right! Renewable sources don't pollute and won't run out. That's why we should use them more."

1.5 Why Should We Save Electricity?

Anaya asked, "But why do we need to save electricity if we can make more?"

"Great question! Here's why:

1. **Saves Money:** Using less electricity means lower bills.
2. **Protects the Environment:** Burning coal and gas pollutes the air.
3. **Saves Resources:** Coal and gas will run out one day.
4. **Reduces Global Warming:** Less pollution means a healthier planet."



1.6 How Can We Save Electricity?

Ms. Priya asked, "Can you think of ways to save electricity at home?"

Maya said, "Turn off lights and fans when not needed."

Aarav added, "Use LED bulbs instead of old bulbs."

Sam said, "Use solar panels!"

Anaya nodded. "And unplug chargers when they are not in use."

"Excellent! Saving electricity is just as important as making it. Let's all promise to use electricity wisely!" Ms. Priya said.

The students cheered. "Yes, we will!"

And that day, they not only learned how electricity is made but also how they could help save it!

1.7 Understanding Electricity and Its Uses

It was a bright Monday morning, and Ms. Priya walked into the classroom with a box. Aarav, Maya, Sam, and Anaya were curious.

"What's in the box, ma'am?" Aarav asked eagerly.

Ms. Priya smiled. "This box holds something magical that powers our world—electricity! Today, we'll explore how we use electricity, how to measure it, and how to be safe while using it."

1.7.1 How is Electricity Used?

Ms. Priya started the lesson by asking, "Can you imagine a day without electricity?"

"No way!" said Sam. "We need it for lights, fans, and even charging our phones."

"Exactly! Electricity is everywhere. It helps us cook, keeps our food fresh in the refrigerator, and even runs our televisions and computers," Ms. Priya said.



Maya added, "Hospitals also need electricity for X-ray machines and ventilators!"

"Great observation, Maya!" Ms. Priya said. "Now, let's see where we use electricity daily."

1.7.2 Common Uses of Electricity in Our Homes

- **Lighting:** Bulbs, tube lights, and LEDs help us see in the dark.
- **Heating:** Electric irons, heaters, and geysers provide warmth.
- **Cooling:** Fans, air conditioners, and coolers keep us comfortable.
- **Kitchen Appliances:** Refrigerators, microwave ovens, and mixer grinders make cooking easy.

1.7.3 Activity 1: Spot the Electricity Users!

Ms. Priya gave the students a task: "Go home and note how many electrical appliances you use in one day. You'll be surprised!"

1.8 Measuring Electricity Consumption

Ms. Priya drew a simple formula on the board:

Energy (kWh) = Power (kW) × Time (hours)

"Let's try an example," she said. "If a 100W bulb runs for 10 hours, how much electricity does it consume?"

Anaya quickly calculated, "100W = 0.1kW, so $0.1 \times 10 = 1$ kWh or 1 unit!"

"Correct!" Ms. Priya said. "This is how electricity usage is measured."

1.8.1 Activity 2: Find Your Home's Energy Use!

"Check your home's electricity bill and find out how many units of electricity your family uses in a month," Ms. Priya said. "Also, see how much is charged per unit!"



1.9 Saving Electricity: Why and How?

"Did you know? Saving electricity not only reduces bills but also helps the environment," Ms. Priya explained. "Here are some easy ways to save energy:"

- Turn off lights and fans when leaving a room.
- Use LED bulbs instead of older ones—they save more power!
- Set AC temperature to 24-26°C instead of 18°C.
- Use natural light during the day instead of switching on lights.

"Let's pledge to save electricity!" Sam suggested. Everyone agreed enthusiastically.

1.10 Staying Safe with Electricity

"Electricity is useful but can be dangerous if not handled properly," Ms. Priya warned. "Let's learn some safety rules."

- Never touch electrical appliances with wet hands.
- Don't overload sockets with too many plugs.
- Keep wires properly insulated.
- Call an electrician for repairs—don't try fixing wires yourself.
- Keep electrical devices away from water.

1.10.1 Activity 3: Spot Safety Signs!

"Look around your school and home for safety signs related to electricity and draw them in your notebook," Ms. Priya instructed.

1.11 The Mystery of the Flickering Lights

One evening, Aarav, Maya, Sam, and Anaya were sitting in their classroom, waiting for Ms. Priya to arrive. Suddenly, the lights flickered for a moment. "Why does that happen?" Aarav wondered aloud.

"Good question!" said Ms. Priya as she walked in. "To understand that, we need to explore three important concepts—voltage, current, and resistance."



1.11.1 Voltage - The Push Behind Electricity

Ms. Priya pulled out a water bottle and poked a hole at the bottom. Water started trickling out.

"Imagine this bottle is like an electrical circuit," she said. "The water inside is like an electric charge, and the pressure that pushes it out is like voltage."

"So, voltage is what pushes electric charges through a wire?" asked Maya.

"Exactly!" Ms. Priya nodded. "Voltage is measured in volts (V) and represented by the letter V. The more voltage, the stronger the push."

She then took out a battery. "Different batteries provide different voltage levels. Let's compare them."

Aarav, curious as always, leaned in. "What's the voltage of this battery?"

"This is an AA **battery**, and it provides **1.5V**," Ms. Priya explained. "A car battery, on the other hand, provides **12V**, and the power in our homes is usually **120V or 230V**."

"Wow! So, higher voltage means more power?" Sam asked.

"Not exactly power, but it does mean a stronger push for electric charges," Anaya clarified.

"If the voltage is too low, the electricity might not be strong enough to power a device."

"That's why some devices need bigger batteries or plug into the wall!" Maya realized.

"Right!" Ms. Priya smiled. "Voltage is like the force behind electricity, making sure everything keeps running."

- AA, AAA batteries – **1.5V**
- Car battery – **12V**
- Mobile phone battery – **3.7V to 4.2V**
- Household voltage (India) – **230V AC**



1.11.2 Tasks for Voltage

- 1.**Conceptual Task:** Explain voltage in your own words using a water pipe analogy.
- 2.**Research Task:** Find out the voltage ratings of five other common electronic devices.
- 3.**Observation Task:** Check the voltage rating on your phone charger and compare it with other chargers at home.
- 4.**Creative Task:** Create a simple infographic explaining different voltage levels used in daily life.

1.12 Current - The Flow of Electricity

Ms. Priya smiled and pointed to a garden hose. "Imagine you're watering plants with a hose," she said.

"The **water pressure** pushing the water out is like **voltage**."

"The **amount of water flowing** through the hose is like **electric current**."

"If the hose is wide, more water flows—just like a thick wire can carry more current."

"If you increase the pressure (voltage), more water flows faster—just like higher voltage pushes more electric charges through a wire!"

Sam's eyes lit up. "So, a thin straw would let out only a little water, like a weak current, but a big pipe would let out a lot, like a strong current?"

"Exactly!" Ms. Priya nodded. "Electric current is just the flow of charge, just like water flowing in a pipe!"

She gave some real-life examples:

- LED lights – Use around **10-20mA**
- Mobile phone charging – Typically **1A to 3A**
- Household appliances – Draw between **5A to 15A**
- Electric heaters and ACs – Can draw more than **20A**

"Current can be dangerous if it's too high," Ms. Priya continued.

- **Less than 1mA** – Barely noticeable



- **1mA - 5mA** – Slight tingling
- **10mA - 20mA** – Muscle contractions
- **30mA - 50mA** – Dangerous
- **Above 100mA** – Can be fatal

1.12.1 Tasks for Current

1 .Conceptual Task: Explain electric current using a water flow analogy.

2 Research Task: Find out the current ratings of five household devices.

3.Observation Task: Check the current rating on your phone charger and compare it with other chargers at home.

4.Safety

Task: Write a short note on why high current levels can be dangerous and how circuit breakers help.

5.Creative Task: Design a table showing different current levels and their effects on the human body.

1.13 Resistance - The Opposition to Current Flow

"Now, what if we placed stones in the water pipe?" Ms. Priya asked.

"The water flow would slow down!" said Anaya.

"Exactly! In an electrical circuit, this slowing effect is called **resistance**. It is measured in **ohms (Ω)** and represented by the letter **R**. The more resistance, the harder it is for current to flow."

She explained that different materials have different resistance levels:

- **Low resistance:** Copper, silver, aluminum (good conductors)
- **High resistance:** Rubber, wood, plastic (insulators)



1.13.1 Color Coding of Resistors

Resistors use a color-coded system to indicate their values:

Color	Digit	Multiplier	Tolerance
Black	0	$\times 1$	-
Brown	1	$\times 10$	$\pm 1\%$
Red	2	$\times 100$	$\pm 2\%$
Orange	3	$\times 1,000$	-
Yellow	4	$\times 10,000$	-
Green	5	$\times 100,000$	$\pm 0.5\%$
Blue	6	$\times 1,000,000$	$\pm 0.25\%$
Violet	7	$\times 10,000,000$	$\pm 0.1\%$
Gray	8	-	$\pm 0.05\%$
White	9	-	-

How to Calculate Resistor Value (for a 4-band resistor)

Example1:

Say the resistor has these color bands:

Red, Violet, Yellow, Gold

Using your table:

Color	Digit	Multiplier	Tolerance
Red	2	-	-
Violet	7	-	-



Yellow	-	$\times 10,000$	-
Gold	-	-	$\pm 5\%$

Calculation:

- First digit = 2
- Second digit = 7
- Multiplier = $\times 10,000$

Value = $(27 \times 10,000) = 270,000 \Omega$ (270 k Ω)

Tolerance = $\pm 5\%$

Example 2:

Colors: Orange, Orange, Brown, Gold

Color	Digit	Multiplier
Orange	3	-
Orange	3	-
Brown	-	$\times 10$

Value = $(3\ 3) \times 10 = 330 \Omega$

Tolerance = $\pm 5\%$



1.13.2 Tasks for Resistance

1. Conceptual Task: Explain resistance using a water pipe analogy. 2

Research Task: Find three materials with high resistance and three with low resistance.

3. Observation Task: Identify three electronic devices at home that use resistors.

4. Color Code Task:

- Decode the resistance value of a resistor with **Red-Red-Brown-Gold** bands.
- Write the color code for a **4.7kΩ** resistor.

5. Calculation Task:

- A circuit has **12V** and **3A** current. What is the resistance?
- A device has **50Ω** resistance and **0.5A** current. What voltage is needed?

6. Creative Task: Draw a resistor color code chart and label each color's meaning.

1.14 Ohm's Law - The Relationship Between Voltage, Current, and Resistance

"All these concepts are connected through **Ohm's Law**," Ms. Priya explained.

$$V = I \times R$$

She showed them some examples:

- If **V = 12V** and **R = 6Ω**, then **I = 2A**.
- If **V = 24V** and **I = 4A**, then **R = 6Ω**.

1.14.1 Tasks for Ohm's Law

1. Conceptual Task: Explain Ohm's Law using a water flow analogy.

2. Research Task: Find out how Ohm's Law is applied in mobile chargers or LED lights.

3. Observation Task: Identify any electrical device at home with voltage and current ratings and calculate its resistance.

4. Calculation Task:

- A **9V battery** is connected to a **3Ω resistor**. Find the current.
- A circuit has **0.5A** current and **20Ω** resistance. Find the voltage.



-
- A heater operates at **230V** and draws **10A**. Find its resistance.

5.Creative Task: Create a step-by-step flowchart to solve Ohm's Law problems easily.

"Electricity is all around us!" Ms. Priya smiled. "Now that you know how it works, you can understand and use it wisely." The students were excited—they had discovered the hidden science behind the flickering lights!

1.15 The Mystery of Electric Currents

Introduction It was a bright morning in Ms. Priya's classroom, and the students were buzzing with excitement. Aarav, Maya, Sam, and Anaya were gathered around the teacher's desk, eager to learn about something new.

Ms. Priya smiled, "Today, we are going to explore the two types of electric currents that power our world: Direct Current (DC) and Alternating Current (AC)."

1.15.1 Direct Current (DC)

"What is Direct Current?" Aarav asked curiously.

Ms. Priya explained, "DC electricity flows in one direction, like water flowing steadily in a straight pipe."

Maya added, "That must be why batteries have a positive and a negative side—they push electricity in a single direction."

"Exactly!" Ms. Priya nodded. "DC is commonly found in batteries, solar panels, and electronic devices."

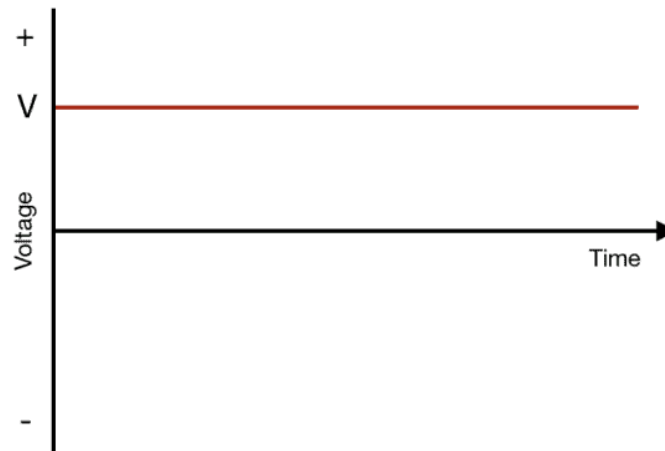
Real-Life Examples of DC:

- **Batteries** – Used in remote controls, mobile phones, and torches.
- **Solar Cells** – Convert sunlight into DC electricity to power homes.
- **USB Devices** – Mobile chargers, power banks, and laptops use DC power.

"Did you know electric cars run on DC?" Sam asked excitedly. "But they need a special AC-to-DC converter to charge their batteries from the power grid."



Diagram: DC Current Flow



The voltage stays constant (no up and down movement). This is why batteries have +(positive) and - (negative) terminals.

1.15.2 Alternating Current (AC)

“What about AC?” Anaya asked.

Ms. Priya smiled, “AC electricity constantly changes direction—it moves back and forth.”

“In India, AC changes direction 50 times per second,” Maya added. “That means it has a frequency of 50 Hz.”

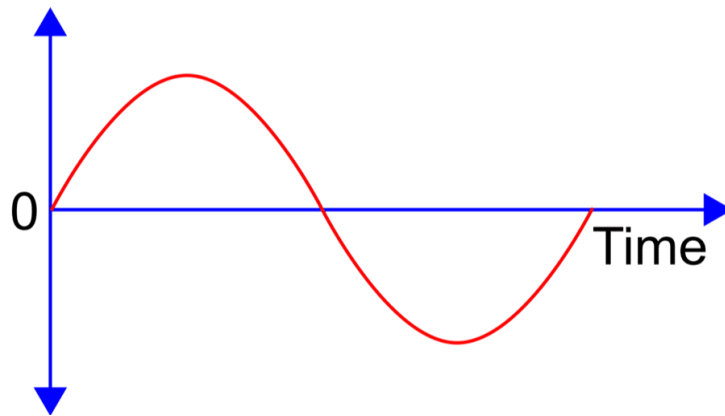
Real-Life Examples of AC:

- **Household Power Supply** – Electricity from sockets is AC.
- **Fans, Refrigerators, and TVs** – All use AC power.
- **Electric Trains & Large Motors** – AC is used for high-power machines.

“Why do we use AC for our homes?” Aarav wondered.

“Great question,” Ms. Priya said. “AC is easier to transmit over long distances without losing much energy.”

Diagram: AC Current Flow



In AC, electricity moves up and down in a wave pattern (sine wave). The voltage rises and falls continuously, changing direction.

1.15.3 Basic Differences Between AC and DC

Feature	Direct Current (DC)	Alternating Current (AC)
Direction of flow	Flows in one direction	Changes direction 50 times per second (50Hz in India)
Source	Batteries, solar cells, USB devices	Power stations, wall sockets
Use	Phones, laptops, cars, LED lights	Household electricity, appliances, factories
Transmission	Short distances only	Long distances (power lines)
Waveform	Straight line	Sine wave

1.15.4 Applications of AC and DC

“When is DC used?” Sam asked.

Ms. Priya replied, “DC is used in: electronic devices (mobiles, laptops, remote controls) Battery-powered gadgets (toys, torches, electric vehicles) Solar energy systems (stores electricity in batteries)”

“What about AC?” Maya wondered.



“AC is used in: Homes and buildings (lights, fans, TVs, washing machines) Factories and industries (large motors and machines) Electric trains and metro systems”

1.15.5 Why Do We Use Both AC and DC?

“Why not use just one type of current everywhere?” Aarav asked.

“Good question,” Ms. Priya said. “AC is better for: Powering homes because it can be sent over long distances without losing much energy.”

“And DC is better for: electronic devices because it provides a steady flow of power.”

“Most home appliances like laptops and TVs use DC, but homes get AC power,” Anaya said. “That’s why they have adapters to convert AC to DC!”

1.15.6 Activity: Identify AC and DC Devices

Ms. Priya handed out an activity sheet and said, “Let’s see if you can identify AC and DC devices in your home.”

Task 1: Look around your home and list 5 devices that use DC power (Hint: Look for battery-operated devices).

Task 2: Identify 3 appliances that use AC power from a wall socket.

Task 3: Observe a mobile charger adapter and check if it says "Input: AC, Output: DC" (it converts AC to DC).

The students eagerly noted down their observations, excited to explore the world of electricity around them.

“Great work, everyone!” Ms. Priya said with a smile. “Electricity powers our lives in amazing ways, and now you know how AC and DC make it all possible.”



1.16 The Power of Electricity

The classroom was buzzing with excitement as Ms. Priya introduced the next topic. "Today, we're going to learn about **power** in electricity," she announced. Aarav, Maya, Sam, and Anaya were ready to dive into another adventure in the world of science.

1.16.1 Definition and Units (Watts)

"Power is the rate at which energy is used or transferred in an electrical system," Ms. Priya began. "It determines how quickly a device consumes electricity to function. The unit of power is the **Watt (W)**."

Maya thought for a moment and said, "So, if a device has higher power, it uses more energy per second?"

"Exactly!" Ms. Priya nodded. The formula for power is:

Power (P) = Voltage (V) × Current (I).

This means the more voltage or current, the higher the power.

Example Calculation: "If a light bulb operates at **220V** and draws **0.5A**, the power consumed is:

$$P = 220V \times 0.5A = 110W$$

This means the bulb uses 110 Joules of energy per second."

Task 1.16.2: Identify Power Ratings

The students were given a task: find the power ratings of five electrical devices at home. Aarav started noting them down in a table:

Device	Power Rating (W)	Usage Time per Day (hours)
TV	100W	3 hours
Fan	75W	6 hours
Laptop	60W	4 hours

Sam observed, "I think the refrigerator consumes the most power because it runs all day."



1.17 Power Calculation ($V \times I$)

Ms. Priya explained, "To understand power better, let's look at some real-life examples."

Case 1: Phone Charger

Voltage: 5V

Current: 2A

Power: $P = 5V \times 2A = 10W$

Case 2: Electric Kettle

Voltage: 220V

Current: 5A

Power: $P = 220V \times 5A = 1100W$

Case 3: Laptop Charger

Voltage: 19V

Current: 3A

Power: $P = 19V \times 3A = 57W$

Task 1.17.1: Calculate Power Consumption

Anaya's eyes lit up. "Let me try! A hair dryer operates at **220V** and draws **7A**."

So, P

$= 220V \times 7A = 1540W$."

"Great! Now, how much energy does it use in **1 hour**?" Ms. Priya asked.

Anaya quickly calculated:

Energy (Wh) = Power (W) $\times$ Time (hours) = $1540W \times 1h = 1540Wh$ or 1.54 kWh.



1.18 Understanding Power Ratings

Ms. Priya pointed to the power ratings on different devices:

Device	Power Rating (Watts)	Usage
LED Bulb	10W	Low
Phone Charger	10W – 20W	Low
Laptop	60W – 100W	Medium
Refrigerator	100W – 800W	High
Microwave Oven	800W – 1500W	Very High
Iron	1000W – 2000W	Very High

"Now, can you tell me which devices consume the most power?" she asked.

"Microwave ovens and irons!" Maya replied.

Task 1.18.1: Power Saving Tips

"How can we save electricity?" Ms. Priya asked.

Sam wrote down three tips:

1. Turn off lights and fans when not in use.
2. Use energy-efficient appliances like LED bulbs.
3. Unplug chargers when devices are fully charged.



1.19 Real-Life Examples of Power Usage

1. Light Bulbs

A **60W incandescent bulb** consumes more electricity than a **10W LED bulb**. Using LEDs saves energy.

2. Phone Chargers

A **10W charger** charges slowly, while a **20W charger** is faster. Higher wattage chargers deliver more power to the phone in less time.

3. Home Appliances

A **refrigerator** runs continuously but consumes less power per second than a microwave. However, a **microwave oven (1000W - 1500W)** consumes a lot of electricity but runs for short durations.

As the class ended, Ms. Priya smiled. "Now that you understand power, next time you plug in a device, you'll know exactly how much electricity it's using!"

The students left, eager to check the power ratings of their home appliances and discover ways to save electricity. Their journey of understanding electricity continued!

1.20 Forward and Reverse Bias

It was another exciting day for Ms. Priya's classroom. Aarav, Maya, Sam, and Anaya were eagerly waiting for today's lesson. Ms. Priya walked in with an electronic circuit board in her hand.

"Today, we'll explore a special component called a diode," she announced.

1.20.1 Basic Semiconductor Concepts

Aarav's eyes lit up. "I've heard about semiconductors, but what exactly are they?" he asked.

Ms. Priya smiled. "Great question! A Semiconductor is a material that is neither a perfect conductor nor a perfect insulator. One of the most common Semiconductor devices is the diode."



She picked up a tiny component from her circuit board. “A diode allows electricity to flow in only one direction.

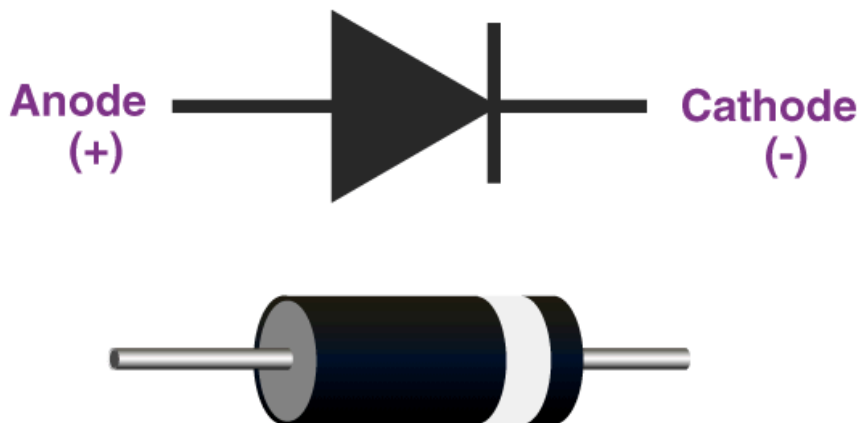
It has two terminals:

1. Anode (+): The positive side
2. Cathode (-): The negative side

Maya leaned forward. “So, it’s like a one-way street for electricity?”

“Exactly!” Ms. Priya nodded. “Diodes are used in many devices, like chargers, LED bulbs, and televisions.”

Diagram: Basic Structure of a Diode



Task: Look around your home and identify devices that might contain diodes.

1.21 How a Diode Works

Ms. Priya continued, “Diodes work in two different modes depending on how they are connected in a circuit. These are called forward bias and reverse bias.”

1.21.1 Forward Bias (Allows Current Flow)

Sam was curious. “What happens in forward bias?”

Ms. Priya explained, “When the anode is connected to the positive terminal and the cathode to the negative terminal, the diode allows current to flow. This is called forward bias.”



She drew a diagram on the board, showing the current flowing through a diode in forward bias.

Real-Life Examples of Forward Bias:

LED Bulbs – LEDs glow only in forward bias.

Mobile Chargers – Diodes help direct current flow while charging.

Electronic Doorbells – The circuit works when the button is pressed.

Maya grabbed a LED and a battery. “If I connect this LED properly, it should glow?”

“Yes!” Ms. Priya said. “Try it and see.”

Task: Use an LED and a battery to create a simple forward bias circuit. Observe when the LED glows.

1.21.2 Reverse Bias (Blocks Current Flow)

Anaya asked, “What if we switch the connections?”

“In that case,” Ms. Priya said, “The diode acts like a closed gate and blocks electricity. This is called reverse bias.”

She drew another diagram showing a diode in reverse bias with no current flow.

Real life of Reverse Bias

Voltage Surge Protection – Prevents appliances from getting damaged by sudden high voltage.

Car Batteries – Stops battery discharge back into the alternator.

Solar Panels – Prevents battery power from flowing back at night.

Anaya nodded. “So, reverse bias is useful for protecting circuits?”

“Absolutely!” Ms. Priya said.

Task: Find an example of reverse bias in an electronic circuit at home (e.g., surge protectors, power adapters).



1.22 Applications of Diodes

1. LED Functioning

Sam held up an LED. “I see LEDs everywhere! How do they work?”

Ms. Priya smiled. “LEDs work in forward bias. When current flows, they emit light.”

Common Uses of LEDs: Traffic lights, TV screens, Car headlights, Mobile phone, flashlights

Task: Check the color and brightness of different LED bulbs at home. Which one is the most efficient?

2. Rectification (Converting AC to DC)

Aarav looked puzzled. “But our homes use AC, and many devices use DC. How does that work?”

Ms. Priya explained, “Diodes are used in rectifiers to convert AC to DC.”

She drew two diagrams:

Half-Wave Rectifier – Uses a single diode to convert AC to DC.

Full-Wave Rectifier – Uses four diodes in a bridge configuration for better efficiency.

Devices that use Rectifiers: Phone chargers, Laptop adapters, Battery-powered devices

Sam exclaimed, “That means my phone charger has a rectifier inside!”

“Exactly!” Ms. Priya said.

Task: Research how mobile chargers use rectifiers to convert AC power to DC.

3. Safety Limits of Diodes

Maya asked, “Can a diode break if too much electricity passes through?”

“Yes,” Ms. Priya said. “Every diode has a voltage limit. If exceeded, it can burn out.”

diodes can handle up to 0.7V, while power diodes can withstand higher voltages.



Importance of Safety limits:

1. Protects electrical appliances from damage.
2. Ensures longer life of diodes and circuits.
3. Helps prevent short circuits.

Task: Check the voltage ratings of different diodes used in electronic devices at home.

Final Task:

Make a simple report on how diodes are used in electronic appliances at home, including LED bulbs, mobile chargers, and voltage protectors.

As the lesson ended, Ms. Priya smiled at her students. “Diodes may be , but they play a big role in our daily lives!”

Aarav, Maya, Sam, and Anaya left the class, excited to find and explore diodes in their home devices!

1.23 Assessment

Read the following MCQ and tick the Correct Answer

1 .Which of the following materials has high resistance?

- ☐ Copper
- ☐ Rubber
- ☐ Aluminum
- ☐ Silver

2 .What is the unit used to measure resistance?

- ☐ Volt
- ☐ Ampere
- ☐ Ohm
- ☐ Watt



3 .In a water pipe analogy, what represents resistance in an electric circuit?

- ☐ Flow of water
- ☐ Pressure of water
- ☐ Stones inside the pipe
- ☐ Size of the pipe

4. Which of the following materials is a good conductor with low resistance?

- ☐ Wood
- ☐ Plastic
- ☐ Copper
- ☐ Rubber

5 . What happens if you increase resistance in a circuit?

- ☐ Current increases
- ☐ Current decreases
- ☐ Nothing changes
- ☐ Circuit becomes more efficient

6. According to Ohm's Law, which formula correctly shows the relationship between voltage, current, and resistance?

- ☐ $V = I \times R$
- ☐ $V = R \div I$
- ☐ $V = I + R$
- ☐ $V = I - R$

7 .Which of the following devices use Direct Current (DC) electricity?

- ☐ Mobile phone
- ☐ Microwave oven
- ☐ Solar panel
- ☐ Electric fan (connected to a wall socket)



8. In which situation is a diode said to be in forward bias?

- ☐ When the anode is connected to the positive terminal and the cathode to the negative terminal
- ☐ When the anode is connected to the negative terminal and the cathode to the positive terminal
- ☐ When no current flows through the diode
- ☐ When the diode is used to stop current in a surge protector

9. What is the unit used to measure electrical power?

- ☐ Ampere
- ☐ Volt
- ☐ Ohm
- ☐ Watt

10. Which of the following devices would most likely have a rectifier inside to convert AC to DC?

- ☐ Laptop adapter
- ☐ Mobile charger
- ☐ Television connected directly to a wall socket
- ☐ LED flashlight